Homework 2 - Space Missions & Systems

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1 Problem definition

1.1 Disturbance torque estimation

In order to evaluate the magnitude of the disturbance torque T_{dy} due to solar radiation pressure, let us write the following vectors in the ICRF reference frame.

$$\hat{\mathbf{X}}_{\mathbf{ICRF}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \tag{1}$$

$$\hat{\mathbf{S}} = \begin{bmatrix} \cos\alpha \\ \sin\alpha \\ 0 \end{bmatrix} \tag{2}$$

$$\hat{\mathbf{n}} = \begin{bmatrix} -\sin\alpha \\ \cos\alpha \\ 0 \end{bmatrix} \tag{3}$$

The equivalent solar radiation force acting on the spacecraft $\mathbf{F_{SRP}}$ is given by the relation in Eq. 4, while the arm of the resulting moment $\mathbf{b}$ is given in Eq. 5. The resulting disturbance torque acting on the spacecraft which is directed along the $\hat{\mathbf{Y}}_{\mathbf{B}}$ is computed as in Eq. 31.

$$\mathbf{F_{SRP}} = -\frac{\phi}{c}\left(\frac{1\ \text{AU}}{D_{MPO}}\right)^2 A\,\hat{\mathbf{n}}\cdot\hat{\mathbf{X}}_{\mathbf{ICRF}}\left((1-C_S)\hat{\mathbf{X}}_{\mathbf{ICRF}} + 2C_S(\hat{\mathbf{n}}\cdot\hat{\mathbf{X}}_{\mathbf{ICRF}})\hat{\mathbf{n}}\right) \tag{4}$$

$$\mathbf{b} = b\hat{\mathbf{S}} \tag{5}$$

$$\mathbf{T_d} = \mathbf{b}\times\mathbf{F_{SRP}} = \begin{bmatrix} 0 \\ T_{dy} \\ 0 \end{bmatrix} \Longrightarrow T_{dy} \simeq -4.24\times10^{-4}\text{Nm} \tag{6}$$

1.2 Control torque division among wheels

Using the mounting matrix $M_{3\times4}$, we can see the effect of each momentum generated by the wheels T_i for $i = 1,2,3,4$ on the overall control torque T_c, as we can see in Eq. 7.

$$\begin{bmatrix} T_x \\ T_y \\ T_z \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & \frac{1}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} \begin{bmatrix} T_{w1} \\ T_{w2} \\ T_{w3} \\ T_{w4} \end{bmatrix} \tag{7}$$

Since we are considering only the first three wheels in the attitude control of the spacecraft, we proceed by considering the submatrix $M_{3\times3}$ made up by the first three rows and the first three columns (we do not consider the fourth wheel). In this way if we invert M, we get that given a required control torque T_y we get the ripartition on each wheel as follows:

$$T_{w1} = 0 \tag{8}$$
$$T_{w2} = -T_y \tag{9}$$
$$T_{w3} = T_y \tag{10}$$

1.3 Equations for the system

The overall system can so be modeled as in Eq. 11.

$$\begin{cases} I\ddot{\theta} = T_d + Mc \\ M_c = -K_p(\theta - \theta_g) - K_d(\dot{\theta} - \dot{\theta}_g) \\ \dot{\omega}_{w1} = 0 \\ \dot{\omega}_{w2} = -\frac{Mc}{I_w} \\ \dot{\omega}_{w3} = \frac{Mc}{I_w} \end{cases} \tag{11}$$

where:

- T_d is the disturbance torque along the pitch axis due to solar radiation pressure
- Mc is the control torque required

1.4 K_p and K_d estimation

The choice of the proportional and derivative gains, K_p and K_d, is made using an iterative process. We consider the characteristic equation for a standard second-order system:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0 \tag{12}$$

where:

$$\omega_n^2 = \frac{K_p}{I} \tag{13}$$
$$K_d = 2\zeta\omega_n I = 2\zeta\sqrt{K_p I} \tag{14}$$

The damping ratio $\zeta = 1$ is chosen to achieve a critically damped response, ensuring the fastest possible settling time without introducing overshoots that could saturate the actuators.

The iterative algorithm to evaluate the optimal K_p and K_d gains works as follows:

- The MATLAB function `logspace()` is used to generate a row vector of logarithmically spaced candidate values for K_p.
- At each step, the system dynamics are simulated using the current K_p and the strictly related K_d.
- After the integration, the results are checked against the system constraints: maximum allowable torque (± 0.511 Nm), maximum single-wheel power consumption (< 29 W), and the final pointing accuracy (< 4 arcsec at $t = 300$ s).
- If any physical or performance constraint is violated, the loop discards the current configuration and proceeds to the next candidate value of K_p.
- As soon as a configuration satisfies all the constraints, the loop terminates, yielding the optimal K_p and K_d pair.

The final values given are:

$$\begin{cases} K_p = 2.222996482526196E + 01 \\ K_d = 7.602493574195317E + 02 \end{cases} \tag{15}$$

1.5 Nadir-Pointing Kinematics

To maintain a nadir-pointing attitude in a circular orbit, the spacecraft must rotate at a constant pitch rate equal to the orbital mean motion n. The mean motion is computed based on Mercury's gravitational parameter μ_M and the orbital radius r:

$$n = \sqrt{\frac{\mu_M}{r^3}} \tag{16}$$

Assuming the spacecraft starts at an inertial angle of $0°$ at $t = 0$, the target guidance angle $\theta_g(t)$ and its derivative (the target pitch rate $\dot{\theta}_g(t)$) are defined as linear and constant functions of time, respectively:

$$\theta_g(t) = n \cdot t \tag{17}$$

$$\dot{\theta}_g(t) = n \tag{18}$$

These guidance signals act as the reference tracking states for the attitude control system.

2 Results

1 - Using the first three momentum wheels, design an attitude control system able to bring the spacecraft to the nadir-pointing condition with a tolerance of ± 4 arcsec, within 5 minutes.

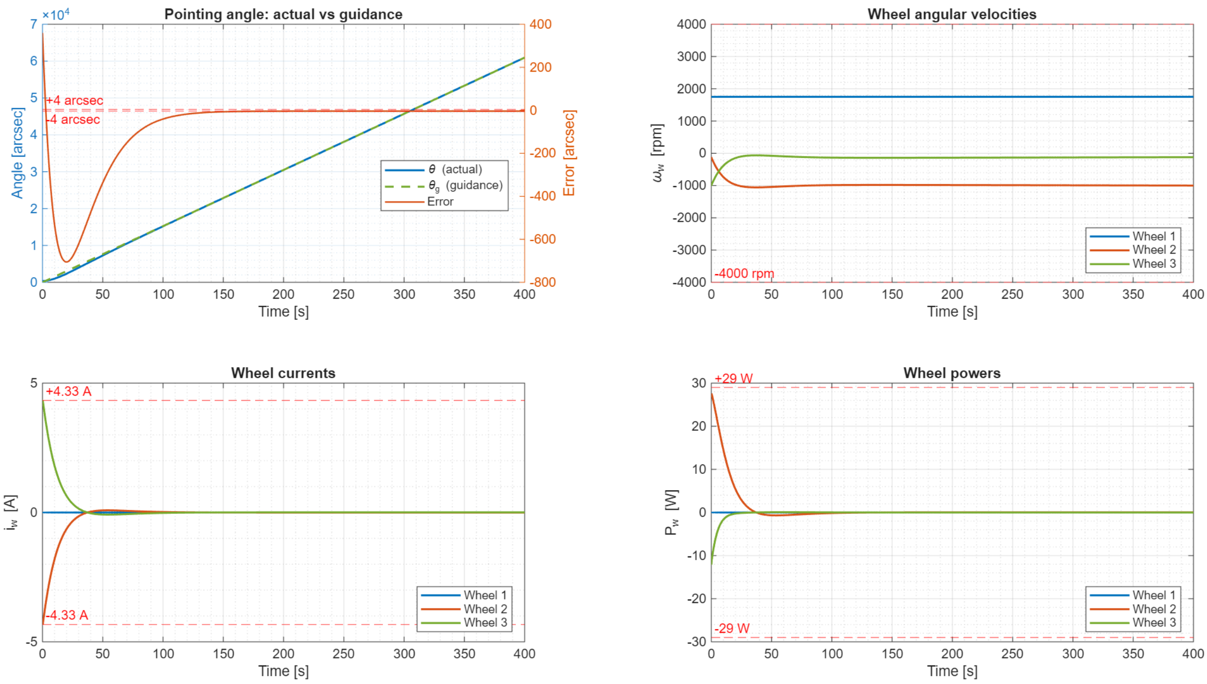

Figure 1: Time evolution of the spacecraft attitude and reaction wheel telemetry during the initial nadir-pointing acquisition maneuver. Top-left: Actual pitch angle versus guidance and resulting pointing error. Top-right: Reaction wheels angular velocities. Bottom-left: Electrical currents drawn by the wheels. Bottom-right: Power consumption per wheel. All physical and performance constraints (dashed red lines) are strictly satisfied.

The performance of the designed Attitude Control System (ACS) during the initial 400 seconds of the simulation is presented in Figure 1.The top-left plot illustrates the spacecraft's pointing kinematics. Starting from the initial offset of $0.1°$ (360 arcsec), the pitch angle θ successfully tracks the linear guidance ramp θ_g dictated by the circular orbit kinematics. The pointing error rapidly converges toward zero, strictly entering and remaining within the required ± 4 arcsec tolerance band well before the 5-minute (300 s) threshold. The remaining three subplots verify the physical constraints of the reaction wheels. The top-right graph confirms that the angular velocities of all active wheels remain safely within the ± 4000 rpm saturation limits. As expected from the inverse mounting matrix, Wheel 1 maintains a constant velocity since no torque is requested along its specific axis for a pure pitch maneuver. Finally, the bottom plots depict the electrical currents and power consumption. The highest demand on the actuators occurs during the initial transient phase to nullify the large initial offset. During this phase, the maximum current drawn remains bounded by the 4.33 A limit, and the peak power consumption of the most stressed wheel safely peaks just below the absolute maximum threshold of 29 W, proving that the chosen PD gains (K_p and K_d) are optimally tuned to maximize performance without violating hardware constraints.

2 - Determine for how long the momentum wheels can maintain the desired nadir-pointing attitude.

After the transient phase the spacecraft reaches the nadir-pointing condition and the control torque along the pitch axis reaches a steady-state condition where

$$Tc = -T_{dy} \tag{19}$$

The reaction wheel angular momentum evolves as

$$\dot{h}_w = -T_c = T_{dy} \tag{20}$$

and, since the disturbance torque T_{dy} is constant, $\dot{h}_w$ evolves linearly at a constant rate. The wheel saturates when its angular momentum reaches the value $h_{sat} = I_w \omega_{max} \cdot sign(T_{dy})$.

If we do not consider an eclipse phase we have that the time interval between the initial condition and the desaturation maneuver is given by

$$t_{sat} = t_{end} + \frac{h_{sat} - h_w(t_{end})}{T_{dy}} \tag{21}$$

where t_{end} is the final time of the simulation in Point 1. In this way we get: $t_{sat2} = 40\,368.63\,\mathrm{s}$ and $t_{sat3} = 52\,090.15\,\mathrm{s}$. From now on for this evaluation we will consider only the saturation of wheel 2 since it is the first one that reaches saturation.

Taking into account the eclipse phase, the enter and exit true anomaly θ_{enter} and θ_{exit} of the eclipse in the considered circular orbits are:

$$\theta_{enter} = \pi - \arcsin\left(\frac{R_M}{r}\right), \qquad \theta_{exit} = \pi + \arcsin\left(\frac{R_M}{r}\right) \tag{22}$$

The eclipse duration and sunlit time per orbit are:

$$t_{ecl} = \frac{\theta_{exit} - \theta_{enter}}{n}, \qquad t_{sunlit} = T_{orb} - t_{ecl} \tag{23}$$

where $T_{orb} = 2\pi/n$ is the orbital period. The angular momentum accumulated per orbit is

$$\Delta h_{orbit} = T_{srp} \cdot t_{sunlit} \tag{24}$$

Starting from t_{end} the total angular momentum budget to saturation is

$$\Delta h_{total} = h_{sat} - h_w(t_{end}) \tag{25}$$

The first partial sunlit phase lasts from t_{end} to the first eclipse entry, contributing:

$$\Delta h_{first} = T_{srp} \cdot (t_{ecl,enter} - t_{end}) \tag{26}$$

If $|\Delta h_{first}| \geq |\Delta h_{total}|$ the wheel saturates before the first eclipse and $t_{sat} = t_{des,no_ecl}$Otherwise, the remaining budget after the first sunlit phase is

$$\Delta h_{rem} = \Delta h_{total} - \Delta h_{first} \tag{27}$$

The number of complete orbits required is

$$N_{orb} = \left\lfloor \frac{|\Delta h_{rem}|}{|\Delta h_{orbit}|} \right\rfloor \tag{28}$$

and the residual angular momentum after N_{orb} complete orbits is

$$\Delta h_{res} = \Delta h_{rem} - N_{orb} \cdot \Delta h_{orbit} \tag{29}$$

The total saturation time is then

$$t_{sat} = t_{end} + (t_{ecl,enter} - t_{end}) + t_{ecl} + N_{orb} \cdot T_{orb} + \frac{\Delta h_{res}}{T_{srp}} \tag{30}$$

which accounts for the first partial sunlit phase, the first eclipse, N_{orb} complete orbits, and the final partial sunlit phase in which saturation occurs.

The following table summarizes the key quantities computed for the desaturation analysis.

Quantity	**Value**	**Units**
Desaturation time without eclipse — Wheel 2	40368.63	s
Desaturation time without eclipse — Wheel 3	52090.15	s
Eclipse entry time	3180.60	s
Eclipse duration	2142.23	s
Desaturation time with eclipse — Wheel 2	53221.99	s
Desaturation time with eclipse — Wheel 2	14.78	h
Number of orbits before saturation	6.26	—

Table 1: Desaturation time analysis for the momentum wheels.

Wheel 2 is the binding constraint, as it saturates first. Since the eclipse entry occurs at $t = 3180.60\,\mathrm{s}$, well before the no-eclipse desaturation time of $t = 40\,368.63\,\mathrm{s}$, the eclipse phase significantly delays the saturation. The momentum wheels can therefore maintain the desired nadir-pointing attitude for approximately 14.78 hours, corresponding to 6.26 orbital periods.

3 - Using the same initial conditions and same geometry, repeat points 1 and 2 assuming now that the orientation of the solar panel orientation is fixed in the spacecraft body frame, with an offset of $-100\,\mathrm{deg}$ with respect to the $\hat{\mathbf{X}}_\mathbf{B}$ axis. .

In this case the spacecraft is subjected by a variable disturbance torque due to solar radiation pressure. It can be modeled as:

$$\mathbf{T_d} = \mathbf{b} \times \mathbf{F_{SRP}} = \begin{bmatrix} 0 \\ T_{dy} \\ 0 \end{bmatrix} \tag{31}$$

where both $\mathbf{b}$ and $\mathbf{F_{SRP}}$ are a function of the angle θ.

Since the solar panel is fixed in the body frame with an offset of $-100\,\mathrm{deg}$ with respect to the $\hat{\mathbf{X}}_\mathbf{B}$, the solar panel direction $\hat{\mathbf{S}}$ and its normal $\hat{\mathbf{n}}$ in the ICRF are:

$$\hat{S} = \begin{bmatrix} \cos(\theta + \alpha_{body} + 90°) \\ \sin(\theta + \alpha_{body} + 90°) \\ 0 \end{bmatrix}, \qquad \hat{n} = \begin{bmatrix} -\sin(\theta + \alpha_{body} + 90°) \\ \cos(\theta + \alpha_{body} + 90°) \\ 0 \end{bmatrix} \tag{32}$$

where $\alpha_{body} = -100\,\mathrm{deg}$ is the fixed offset angle in the spacecraft body frame. The moment arm vector is:

$$\mathbf{b} = b\hat{\mathbf{S}} \tag{33}$$

and the SRP force is computed using Eq. 4, yielding a disturbance torque T_{dy} that varies periodically with the orbital angle $\theta(t) = n \cdot t$.

If we integrate over an orbital period T_{orb}, we get that the disturbance torque in this case is the one in Fig. 2 with mean and extremal values in Table 2.

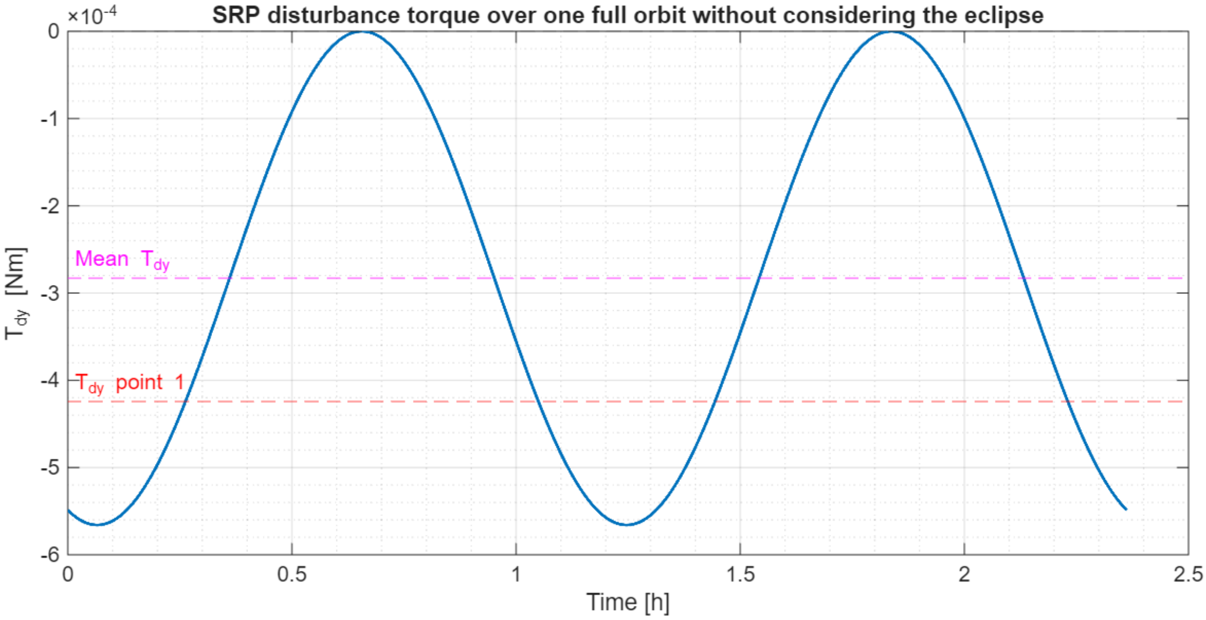

Figure 2: Behavior of the disturbance torque T_{dy} due to solar radiation pressure in the case of a fixed solar panel configuration in the body frame over one full orbital period.

	Value	**Unit**
Mean T_{dy}	-2.8296×10^{-4}	Nm
Max T_{dy}	-3.3946×10^{-35}	Nm
Min T_{dy}	-5.6586×10^{-4}	Nm

Table 2: Solar Radiation Pressure disturbance torque T_{dy} statistics over one orbit.

Using the iterative method of Point 1 the loop fails to find a fit for K_p and K_d that satisfy all the constraints. It is needed in this case a PID control law.

A similar iterative process is implemented using `logspace()` but in this case iterating on the natural frequencies of the system ω_n and using the following relations to find the gains:

$$K_p = 3I\omega_n^2 \tag{34}$$
$$K_d = 3I\omega_n \tag{35}$$
$$Ki = I\omega_n^3 \tag{36}$$

In this iterations the system behavior is simulated on a full orbital period in order to evaluate the response of the ACS when the disturbance module $|T_{dy}|$ is maximum. The loop gives in output the following values for the gains:

$$\begin{cases} K_p = 27.640746 \\ K_d = 734.162482 \\ K_i = 0.346885 \end{cases} \tag{37}$$

In Figure 3 the system behavior in the first 400 s from the initial condition using these values of the gains is plotted.

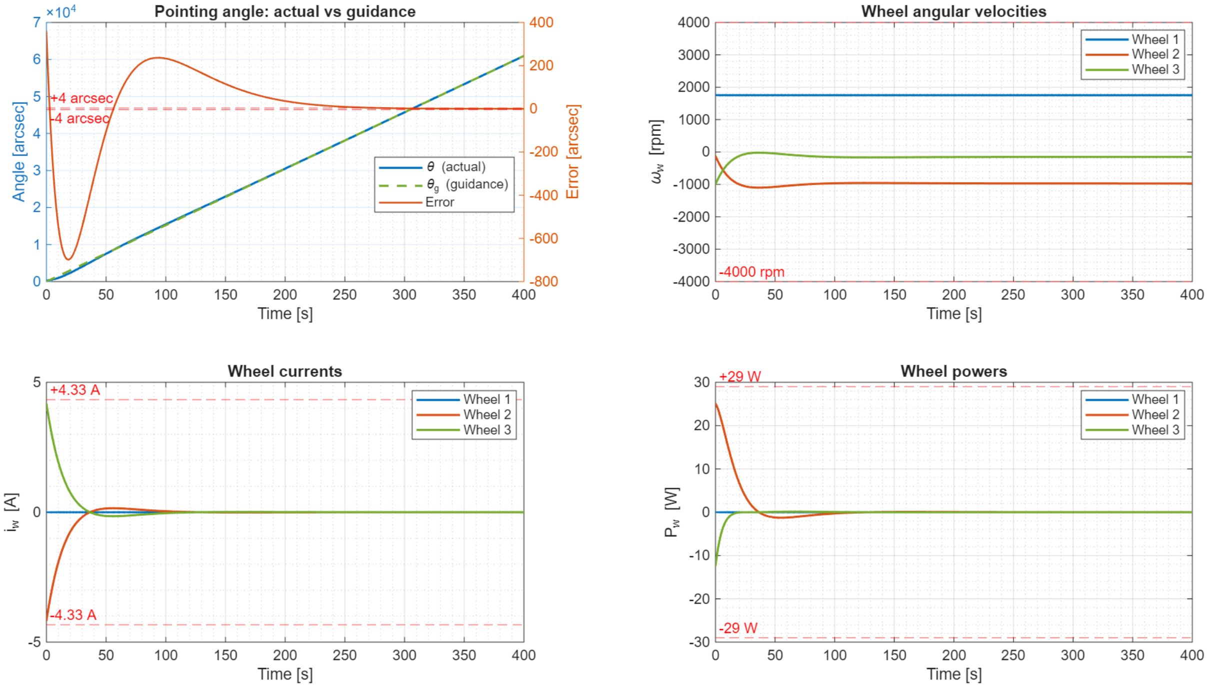

Figure 3: Time evolution of the spacecraft attitude and reaction wheel telemetry during the initial nadir-pointing acquisition maneuver under a variable disturbance torque. Top-left: Actual pitch angle versus guidance and resulting pointing error. Top-right: Reaction wheels angular velocities. Bottom-left: Electrical currents drawn by the wheels. Bottom-right: Power consumption per wheel. All physical and performance constraints (dashed red lines) are strictly satisfied.

To evaluate the wheels' saturation time, the system is simulated over multiple orbital periods until any reaction wheel reaches its maximum angular velocity. Using this approach we get the results in Table 3.

	Value	**Unit**
Saturation time t_{sat}	67082.60	s
Saturation time t_{sat}	18.63	h
Saturation time t_{sat}	7.89	orbits
Wheel 2 angular velocity ω_{w2}	-4000.00	rpm
Wheel 3 angular velocity ω_{w3}	2879.00	rpm

Table 3: Reaction wheels saturation time and final angular velocities.

Wheel 2 is the binding constraint, reaching its maximum angular velocity of -4000 rpm first. Wheel 3, despite accumulating angular momentum in the opposite direction due to the torque distribution ($T_{w3} = T_c = -T_{w2}$), has not yet reached saturation at $\omega_{w3} = 2879$ rpm when the simulation terminates. The momentum wheels can therefore maintain the desired nadir-pointing attitude for approximately 18.63 hours, corresponding to 7.89 orbital periods.